

Topic 02 - Tutorial Sheet-01

First Steps

Exercise 1:

Parenthesise the following numeric expressions:

1. $2^3 * 4$

2. $2 * 3 + 4 * 5$

3. $2 + 3 * 4^5$

Use GHCi to check your solutions.

Solution 1:

1. $(2^3) * 4$

2. $(2 * 3) + (4 * 5)$

3. $2 + (3 * (4^5))$

Exercise 2:

The library function *last* selects the last element of a non-empty list; for example:

$$\text{last } [1,2,3,4,5] = 5$$

Show that the function *last* could be defined in terms of the other library functions introduced so far.

Solution 2:

```
last xs = head (reverse xs)
```

Exercise 3:

The library function *init* removes the last element of a non-empty list; for example:

$$\mathit{init} \ [1,2,3,4,5] = [1,2,3,4]$$

Show that the function *init* could be defined in terms of the other library functions introduced so far.

Solution 3:

```
myinit xs = reverse(tail (reverse xs))
```
